Computer Science - Undergraduate Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following statements about a remote procedure call is true?
   (a) It is used to call procedures with addresses that are farther than $2^1$6 bytes away.
   (b) It cannot return a value.
   (c) It cannot pass parameters by reference.
   (d) It cannot call procedures implemented in a different language.

2. Which is a valid expression in Python 3.5?
   (a) sort(’ab’)
   (b) sorted(’ab’)
   (c) "ab".sort()
   (d) 1/0

3. Let l = [1,2,2,3,4]. In Python 3, what is a possible output of set(l)?
   (a) {1}
   (b) {1,2,2,3,4}
   (c) {1,2,3,4}
   (d) {4,3,2,2,1}

4. Which of the following is not a sentence that is generated by the grammar A -> BC, B -> x|Bx, C -> B|D, D -> y|Ey, E -> z?
   (a) xyz
   (b) xy
   (c) xxzy
   (d) xxxxy

5. Which of the following statements is FALSE about memory reclamation based on reference counting?
   (a) Reference counting is well suited for reclaiming cyclic structures.
   (b) Reference counting incurs additional space overhead for each memory cell.
   (c) Reference counting is an alternative to mark-and-sweep garbage collection.
   (d) Reference counting need not keep track of which cells point to other cells.

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression ['Hi!'] * 4 raises an error due to invalid syntax.
7. True or False: Identifier length is best specified using a context-free grammar in programming languages.
8. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is n!/(n - m)!
9. True or False: Church's thesis was first proven by Alan Turing, establishing a definitive link between computable functions and Turing machines.
10. True or False: Interrupts are used in place of data channels to transfer data between devices and memory.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the given number is a perfect square or not.
12. Write a function to caluclate perimeter of a parallelogram.

End of Paper